

PARAINEN FAGERHOLM, Finland

WMO: 029500

Lat: 60.112N

Lon: 21.698E

Elev: 5

StdP: 101.27

Time Zone: 2.00 (EUE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.0	-12.6	-18.5	0.7	-15.8	-15.3	1.0	-11.5	15.8	0.8	14.2	1.0	3.4	40	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	24.8	20.3	23.0	19.2	21.5	18.1	21.5	23.8	20.0	22.2	18.7	20.8	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.3	22.8	19.2	14.0	21.3	17.9	12.9	19.9	62.6	23.5	57.4	22.3	53.2	20.8	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.5	11.1	-15.7	27.0	6.0	2.5	-20.0	28.8	-23.5	30.2	-26.9	31.6	-31.2	33.4		
(5)			-16.3	21.2	6.2	2.4	-20.7	22.9	-24.3	24.3	-27.8	25.7	-32.3	27.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-2.0	-2.4	-0.4	3.7	9.7	14.1	18.0	17.4	13.5	8.0	4.5	1.3		
	DBStd	7.87	4.37	5.18	3.51	2.53	3.63	2.65	2.79	2.16	2.53	2.81	3.47	4.03		
	HDD10.0	1797	373	346	323	190	51	2	0	0	4	73	165	269		
	HDD18.3	4122	632	579	582	438	267	130	41	44	145	321	415	528		
	CDD10.0	764	0	0	0	1	42	124	249	229	109	10	0	0		
	CDD18.3	49	0	0	0	0	0	2	32	14	1	0	0	0		
	CDH23.3	138	0	0	0	0	1	8	107	22	0	0	0	0		
	CDH26.7	17	0	0	0	0	0	0	15	2	0	0	0	0		
(14) Wind	WSAvg	5.4	5.9	5.4	5.4	4.6	4.5	5.0	4.7	5.0	5.2	5.9	6.6	6.7		
(15) Precipitation	PrecAvg	620	55	40	38	32	27	47	54	66	60	73	71	60		
	PrecMax	759	93	84	87	56	72	79	109	122	187	144	156	99		
	PrecMin	474	9	3	5	5	5	14	4	8	11	33	11	12		
	PrecStd	65	23	22	18	16	19	18	32	30	46	26	37	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.2	7.6	14.4	21.9	24.5	28.6	26.3	20.2	14.2	10.6	8.4		
		MCWB	4.3	3.4	3.7	8.9	14.9	17.4	21.2	21.3	17.9	12.1	9.5	7.0		
	2%	DB	4.6	4.1	5.8	11.3	19.6	21.6	26.3	23.3	18.6	13.1	9.9	7.2		
		MCWB	3.8	3.0	3.6	7.2	14.1	16.4	20.6	19.9	16.3	11.8	9.2	6.1		
	5%	DB	4.0	3.2	4.7	9.1	17.7	20.0	24.1	21.8	17.7	12.3	9.2	6.4		
		MCWB	3.3	2.6	3.0	5.8	13.3	15.6	20.2	18.6	15.6	11.1	8.4	5.4		
	10%	DB	3.2	2.6	3.6	7.4	15.7	18.4	22.6	20.6	16.8	11.6	8.3	5.7		
		MCWB	2.5	2.1	2.2	4.9	11.9	14.8	19.3	17.7	14.9	10.5	7.4	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.2	5.4	9.5	16.6	19.7	23.8	23.2	18.4	13.0	10.0	7.4		
		MCDB	5.0	4.8	6.7	13.4	19.6	22.2	24.7	24.0	19.2	13.4	10.2	8.0		
	2%	WB	4.1	3.2	3.9	7.6	14.8	17.4	22.2	20.7	17.0	12.2	9.2	6.4		
		MCDB	4.4	3.7	5.0	10.4	18.3	20.4	24.9	22.6	18.1	12.7	9.8	6.9		
	5%	WB	3.4	2.7	3.1	6.2	13.5	16.3	20.9	19.4	16.0	11.6	8.5	5.6		
		MCDB	3.9	3.1	4.2	8.4	17.1	19.0	23.3	21.1	17.2	12.1	9.1	6.2		
	10%	WB	2.6	2.1	2.5	5.3	12.2	15.4	19.6	18.4	15.3	10.7	7.6	4.9		
		MCDB	3.1	2.6	3.4	7.0	15.3	17.7	22.0	19.9	16.5	11.4	8.2	5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.8	4.8	5.6	6.1	5.3	5.0	4.1	3.7	3.1	2.6	2.9		
		MCDBR	2.8	3.3	5.1	8.7	9.3	8.3	7.5	5.7	4.5	3.1	2.8	3.0		
	5% WB	MCWBR	2.8	2.8	3.6	5.0	4.9	4.3	3.5	2.9	3.0	2.8	2.7	3.0		
		MCWBR	2.8	3.2	4.4	7.5	8.8	7.2	6.1	4.4	3.9	2.9	2.8	2.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.281	0.315	0.333	0.372	0.368	0.361	0.389	0.380	0.351	0.335	0.307	0.393		
		taud	2.174	2.249	2.309	2.319	2.418	2.478	2.423	2.437	2.502	2.404	2.216	2.637		
	Edn at Noon	Edn at Noon	481	676	793	825	859	872	833	811	778	655	439	338		
		Edh at Noon	42	69	89	105	103	99	102	94	75	60	41	30		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.24	0.82	2.37	4.04	5.55	6.06	5.65	4.41	2.71	1.15	0.32	0.15		
	RadStd	RadStd	0.03	0.15	0.27	0.40	0.49	0.40	0.50	0.39	0.31	0.17	0.07	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	-128	N/A	N/A	N/A

Nomenclature: See separate page